Beacon: Milestone 2 Progress Report

ITCC 41 | Group 12 | Acenas, Bagaporo, Mercado

Date: November 11, 2026 | Repository: https://github.com/IcyAcenas/BeaconApp.git

This report covers the progress of Beacon for Milestone 2, where we built the user interface from the design submitted in Milestone 1. The Home Map is complete and follows the approved mockup. The other screens and the navigation between them are still to be built. The app currently runs as a mobile-sized web page and has not been moved into Apache Cordova yet.

1. What Was Completed

The Home Map is the screen every other screen opens from, so we built it first. It shows a full-screen dark map of Cagayan de Oro with the Beacon logo, profile button, and search bar on top. The hazard pins use the Milestone 1 colors: blue for flash floods, red for accidents, and orange for fires. The landslide pin is green for now because that report is already resolved, while the design uses brown for landslides. Each pin opens a popup when tapped.

The warning banner under the search bar changes its message when a hazard is selected. For now it responds to taps only and does not use the phone's location.

The Active Hazards sheet at the bottom has a card for each category with the photo, reporter, time posted, and an active or resolved status dot. It opens and closes by tapping the handle or title, or by swiping. Selecting a card or a pin highlights the card and moves the map to that hazard, and the search bar filters the cards. We also added a profile panel that opens from the profile button. It was not in the design and is only a placeholder. The layout uses the dark green and cream colors from Milestone 1 and is made for a phone screen.

Screen	Status	Notes
Home Map	Done	Map, top bar, warning banner, and Active Hazards sheet with swipeable cards.
Profile panel	Done (added)	Slide-up panel from the profile button. Placeholder content.
Sign In	Not started	Email and password login or sign-up with driver and responder roles.
Plan Route and Route Map	Not started	From and To fields with suggestions, then the route screen.
Hazard List	Not started	Report list for each category, sorted by time or distance.
Report Hazard	Not started	Form with location, photo or video, and description.
Admin Dashboard	Not started	Incident Triage Queue with filter chips and Mark as Resolved.

Table 1. Status of the screens.

2. Screenshots of Implemented Screens

The screenshots below show the screens built so far.

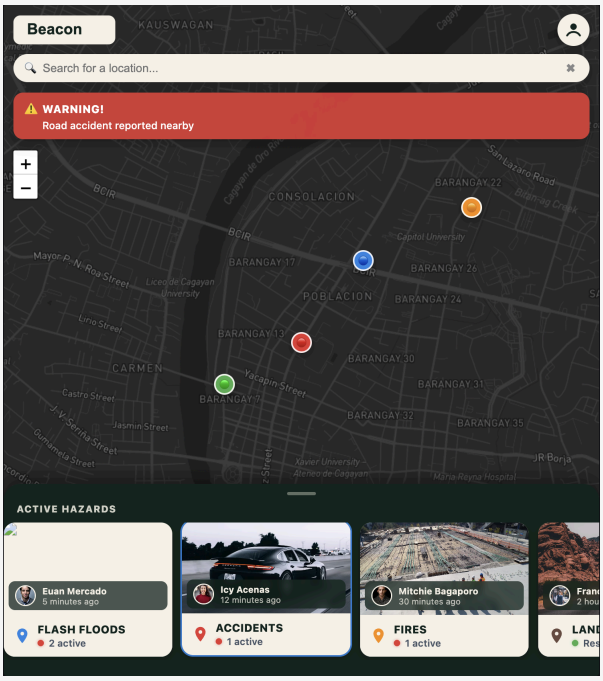


Figure 1. Home Map

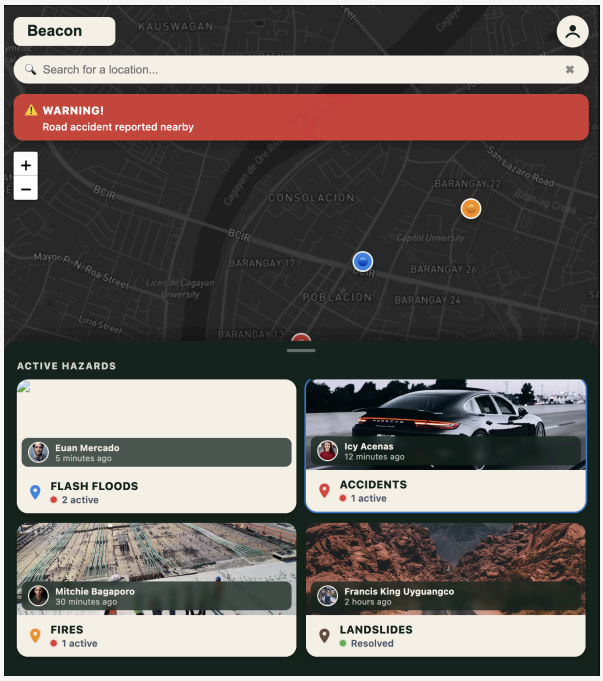


Figure 2. Active Hazards sheet pulled up

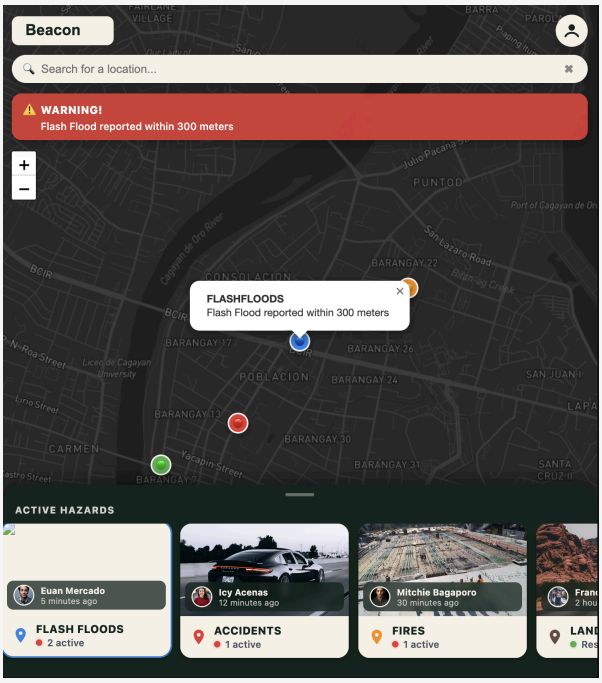


Figure 3. Hazard selected with the warning banner

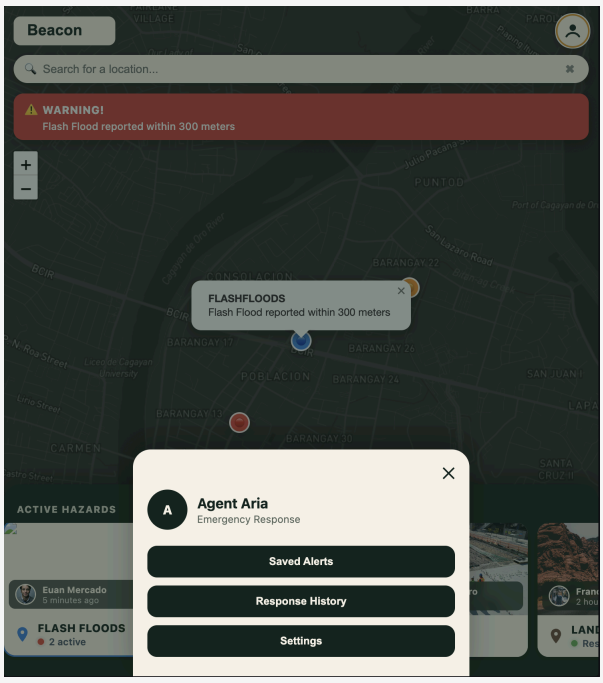


Figure 4. Profile panel

3. What Remains to Be Developed

The following still need to be developed:

- Sign In, Plan Route, Route Map, Hazard List, Report Hazard, and Admin Dashboard screens.
- Navigation between screens, including the back arrows and the Switch Dashboard button for responders.
- Forms and inputs: login and sign-up, the From and To fields, and the report form.
- Firebase for accounts and reports, the phone's GPS, the 300 meter warning, and the Mapbox route. The hazards shown now are sample data.
- Moving the project into Apache Cordova and building the APK.

4. Problems Encountered and How They Were Addressed

Problem	How it was addressed
Getting a live Mapbox map to work in our code was difficult.	We loaded the Mapbox dark style into Leaflet as map tiles using our access token. This is not Mapbox GL JS as stated in Milestone 1, but the map works and hazard pins can be placed on it. The Mapbox Directions API can still be used for the route later.

Problem	How it was addressed
The logo, search bar, and warning banner covered the map zoom buttons.	We moved the zoom buttons lower and fixed the top items in place above the map.
Swiping the bottom sheet also scrolled the hazard cards sideways.	When the sheet is closed, touching the cards only scrolls them. Swiping on the handle or the title opens the sheet.
The final logo, icons, and photos were not available yet.	We used a simple drawn logo and icon and sample photos from the internet. These will be replaced with our own files.

Table 2. Problems encountered and how they were addressed.